Track A · Technology & Engineering | Smart Water & Digital Innovation

AquaReach: Fair and Explainable Emergency Water Distribution for Water Supply Disruptions

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C The problem

- When a pipe bursts or a treatment plant shuts down, utilities send **water tankers and static tanks**. Tankers, water and time are limited.
- Dispatch often follows **who complained first** or **who is closest**, not who will run out first.
- Water at a tank downstairs is not water in every home. **Older residents and persons with disabilities on high floors** cannot carry it up.

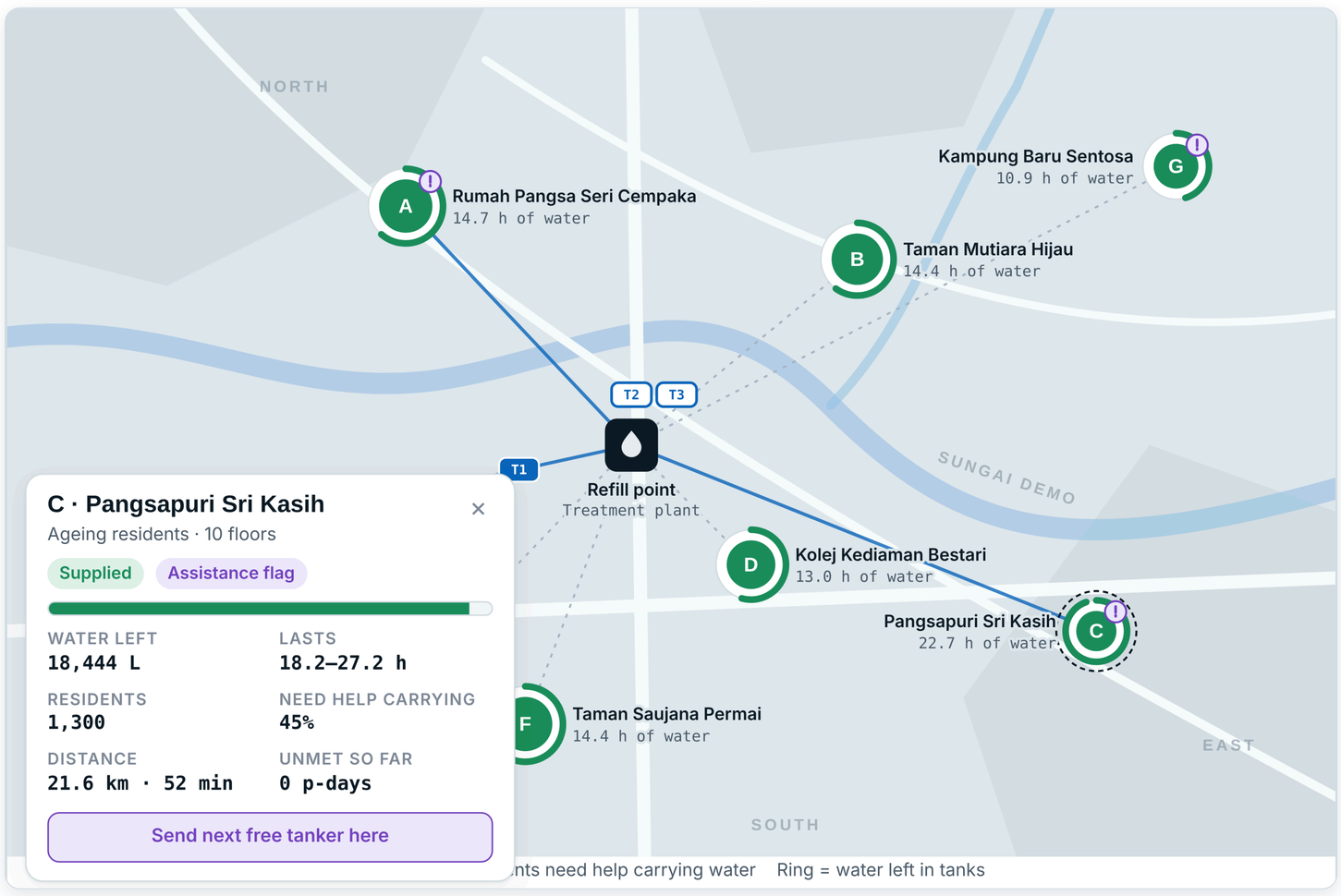
15 L
per person per day: the minimum for drinking, cooking and hygiene in emergencies (Sphere)

D Objectives

1. Estimate how long each community's water will last, and recommend **who to serve, how much, and in what order**.
2. Reduce **unmet essential water** and the **longest time any community goes dry**, compared with common dispatch rules.
3. Explain every recommendation, flag residents who need help collecting water, and keep **dispatchers in control**.

> With the same tankers and the same water, can a transparent method leave fewer people without essential water?

E Our solution: the AquaReach dispatch console



A web-based **decision-support tool for emergency water dispatchers**. It turns scattered information into a ranked, explained delivery plan.

Recommended next dispatches — RANKED BY URGENCY + ACCESS NEEDS

1	**G · Kampung Baru Sentosa** Runs out in 10.0–15.0 h · 22% need help carrying water · 63 min away	Send next
2	**H · Flat Sri Kenanga** Runs out in 11.4–17.1 h · 28% need help carrying water · 44 min away	Send next
3	**A · Rumah Pangsa Seri Cempaka** Runs out in 13.2–19.8 h · 32% need help carrying water · 35 min away	Send next
4	**F · Taman Saujana Permai** Runs out in 10.1–15.2 h · 32 min away	Send next
5	**E · Pangsapuri Delima** Runs out in 12.5–18.7 h · 24% need help carrying water · 45 min away	Send next

1. **Live map** shows hours of water left in every community.
2. **Ranked queue** says why each site comes first.
3. **Override**: the dispatcher always has the final say.
4. **Assistance flags** alert volunteers for door-to-door help.

F How does it work?

Check-in
Building managers report tank levels; population and access needs on file

→ **Estimate**
Hours of water left for each community, as a range

→ **Prioritise**
Rank by time to empty, access needs and travel; respect tanker limits

→ **Approve**
Dispatcher sees the reasons, then approves or overrides

→ **Deliver & assist**
Tankers go out; volunteers help residents who cannot carry water

↻ New readings feed the next cycle, so the plan updates as the situation changes.

G What's innovative

Time-to-empty forecasting
Plans around when water runs out, not when someone complains.

Fair, need-based allocation
Weights communities with more vulnerable residents and less storage.

The last 50 metres
Water at a tank is not water in every home. Assistance flags close the gap.

Transparent by design. Built on established optimisation methods, not a black box. Every decision shows its trade-off, and humans stay in control.

H Expected outcomes & impact

−63% unmet essential water vs first come, first served

−80% unmet essential water vs nearest first

0.8 h longest dry spell, down from 2.6 h and 19.3 h



Simulated, not operational data: mean of 500 synthetic scenarios (8 communities, 3 tankers × 10,000 L, 48 h). The trade-off is 11–16% more travel. The gains held from 38% to 84% across different scarcity levels.

Environmental
Every litre goes where it is needed most; stronger water security during climate-driven disruptions.

Social
Fairer access for older residents, persons with disabilities and low-storage homes; greater public trust.

Economic
More value from existing tankers with no new hardware; fewer complaints and escalations.

I Feasibility & scalability

✓ DONE
Prototype
Simulation tested on 500 scenarios; interactive dispatch console built.

NEXT
Shadow-mode pilot
6–10 sites (hostels, high-rise blocks). Runs alongside real dispatch without controlling it.

SCALE
District to state
Link to utility systems; extend to other states and flood relief.

Needs: a light web app, CSV uploads and open-source solvers. **Partners:** water utilities, local councils, building managers, residents' associations and community volunteers.

J Key message

Same tankers. Same water. Fairer outcomes.

AquaReach makes every emergency litre count, so a sustainable water future also leaves no one behind.

References: Sphere Association (2018). *The Sphere Handbook*, 4th ed. · Toth, P. & Vigo, D. (2014). *Vehicle Routing: Problems, Methods, and Applications*, 2nd ed. SIAM. · UN General Assembly (2010). Resolution A/RES/64/292, The human right to water and sanitation.